FISH HATCHERIES





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FISH HATCHERIES

- Fish hatcheries are used to aid in the breeding, hatching and rearing of fish through the early stages of life.
- They can be run by individual, a cooperative, an institution or a community.

FISH HATCHERIES

Functions of the Fish Hatchery.

1. Acts as a reliable sources of **quality** and **quantity** fingerlings.

What is quality? (healthy, uniform size, high survival rate, strong to with stand transportation stress, Age of between 30 and 120 days)

What is quantity? (Enough and provided consistently) or the supply must meet the demand.

FISH HATCHERIES

Functions of the Fish Hatchery.

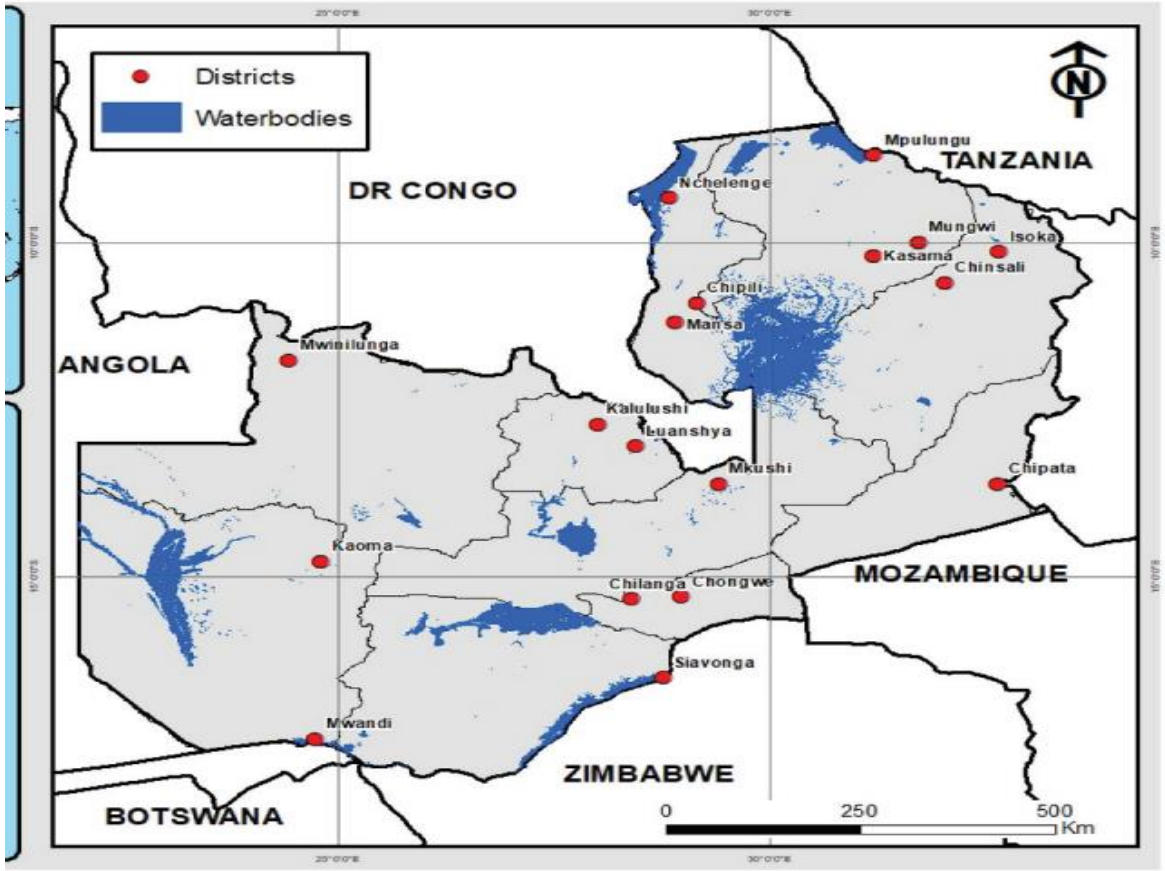
2. Reducing fingerling mortality (due to short distance of transportation)

3. Reducing the amount of money spent on transporting fingerlings

4. Acts a place of improving skills, knowledge and competencies (Point of demonstration)

FISH HATCHERIES

Location of selected government and private hatcheries in Zambia





Cogent Food & Agriculture

Taylor & Francis
Taylor & Francis Group

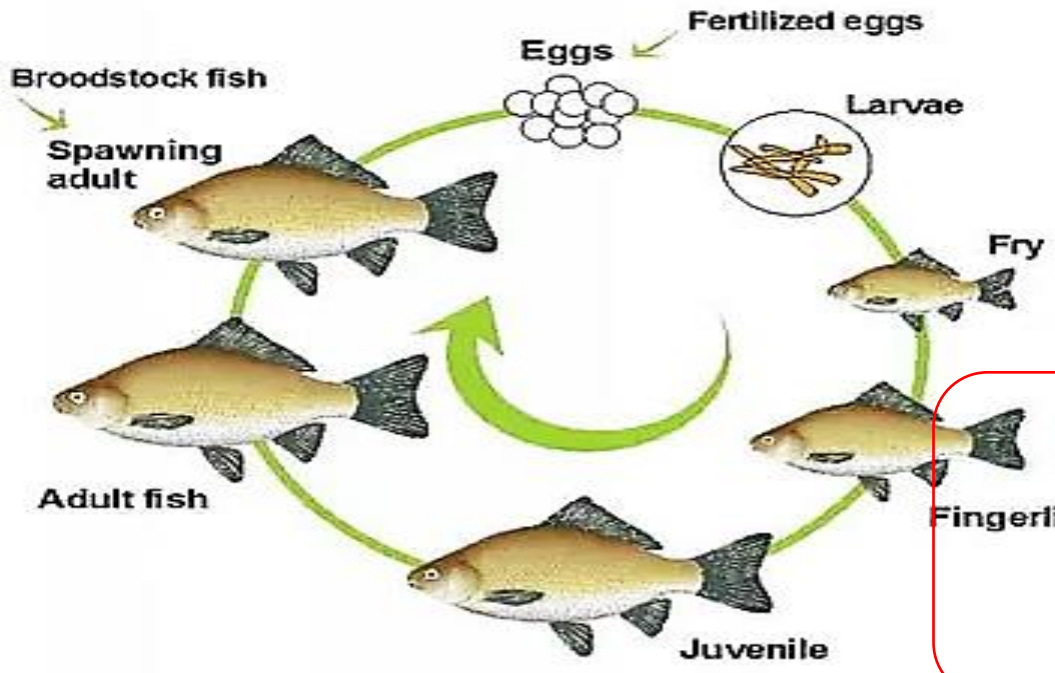
ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/oafa20

Application of anaesthetics in fish hatcheries to promote broodstock and fish seed welfare in Zambia

Joseph Mphande, Oliver J. Hasimuna, Edwin Kikamba, Sahya Maulu, Kundananji Nawanzi, Darius Phiri, Moses Chibesa, Enock Siankwilimba, Chisomo J. Phiri, Buumba M. Hampuwo, Valdemiro Muhala & Sillah Siavwapa

FISH HATCHERIES

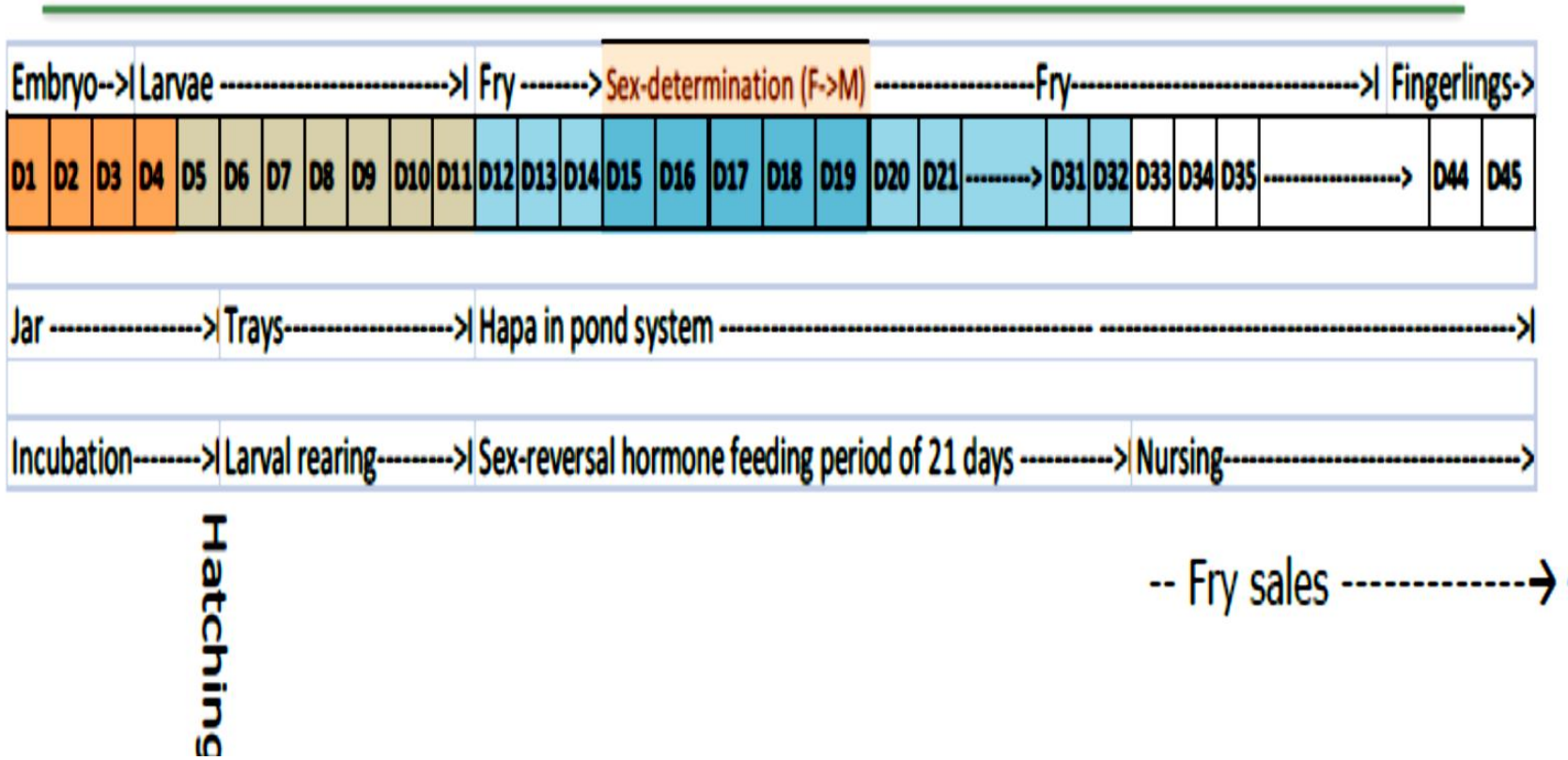
Life cycle of a fish



SEED

FISH HATCHERIES

Life cycle of a fish

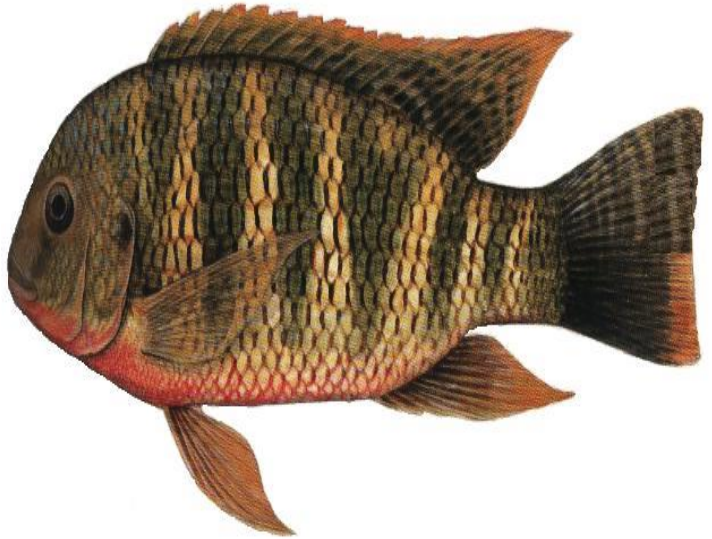


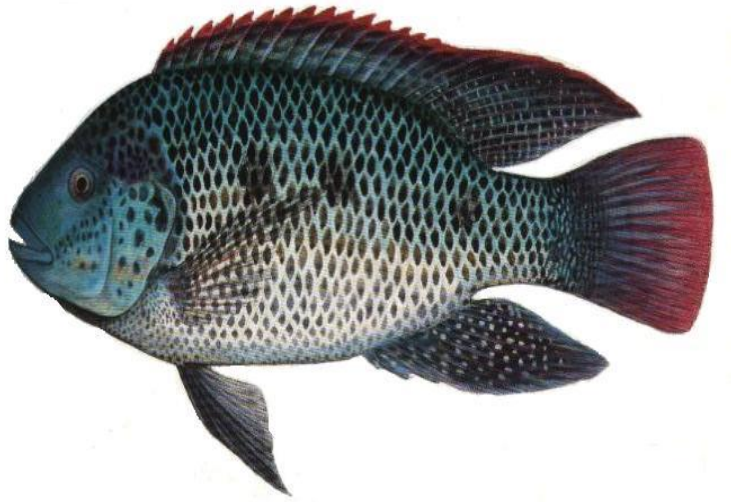
FISH HATCHERIES

DESIGNS OF HATCHERIES

It is critical to know the species for which the hatchery is being designed.

For instance, the breeding behavior of the fish (*C.rendalli and O.andersonii*)

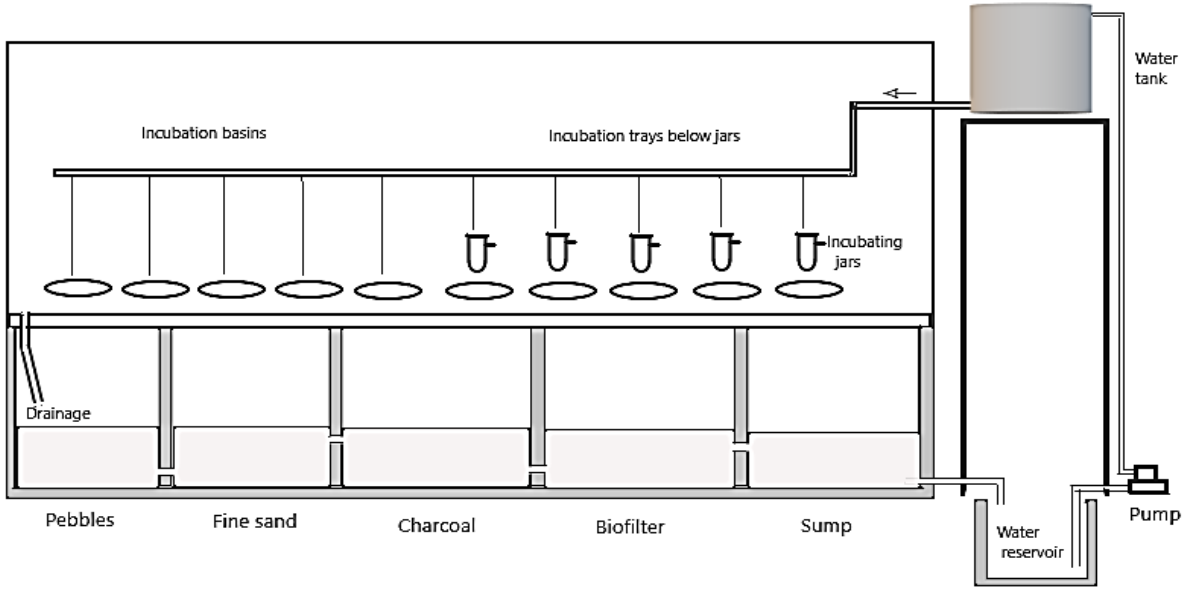


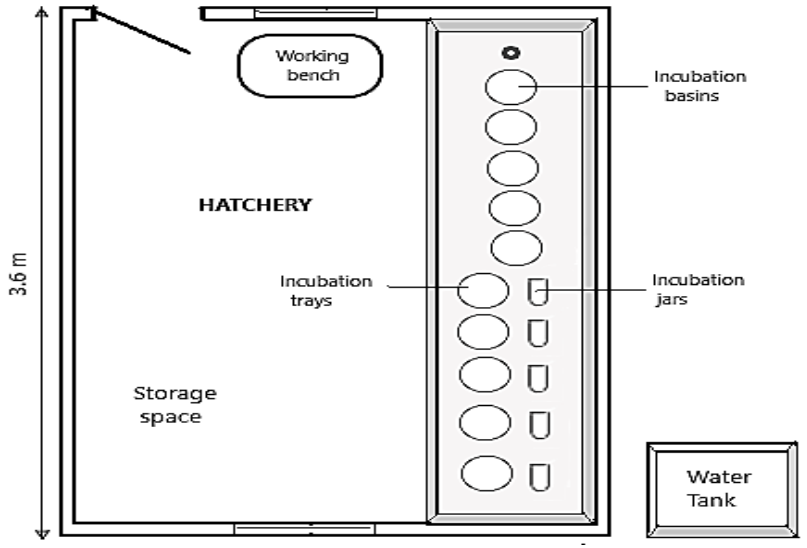


FISH HATCHERIES

DESIGNS OF HATCHERIES

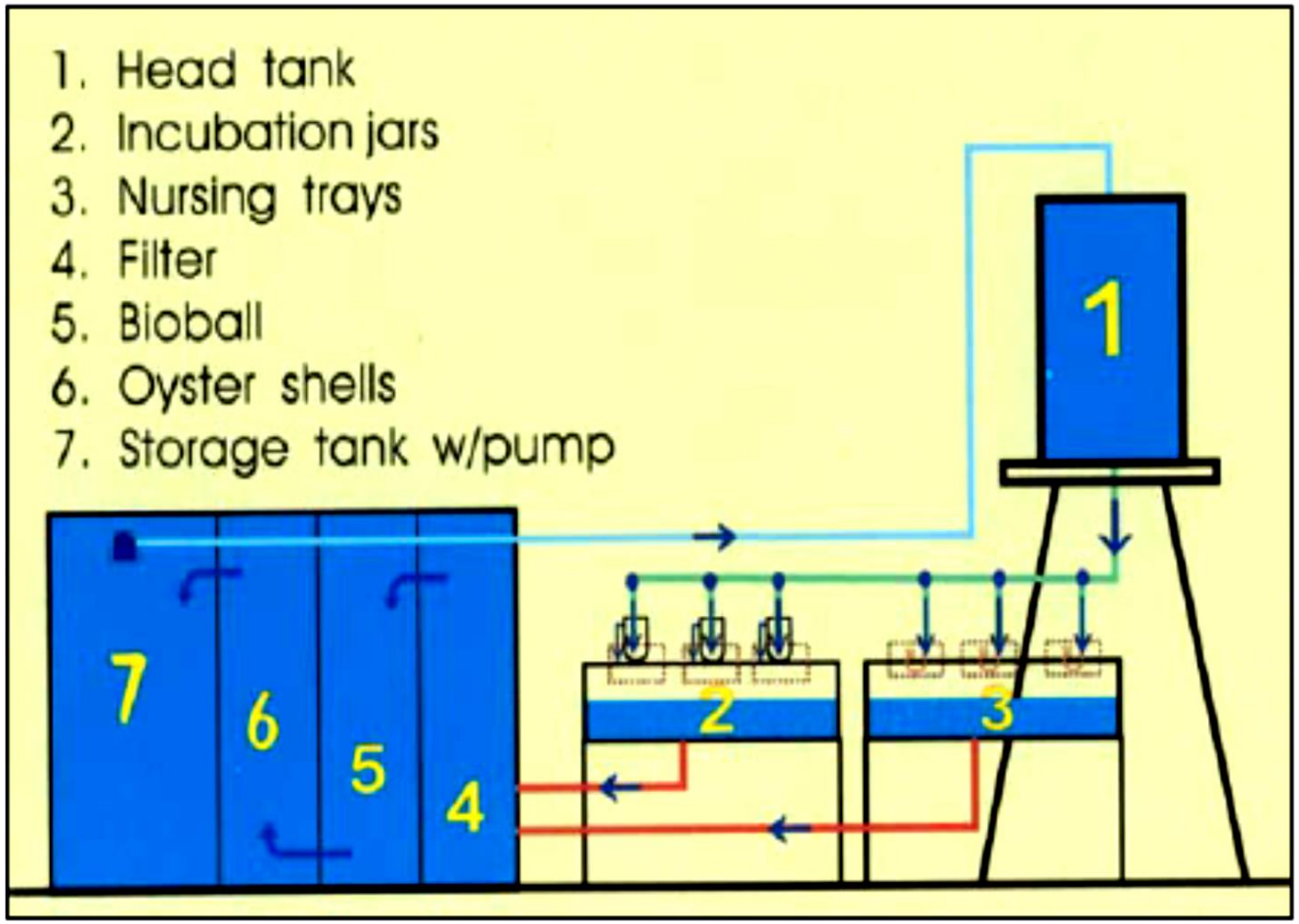
- There are elements common to most hatcheries.
- Bio-security is also very important (Fencing, foot baths at the entrance of hatcheries).





FISH HATCHERIES

DESIGNS OF HATCHERIES



FISH HATCHERIES

SIMPLE LAYOUT

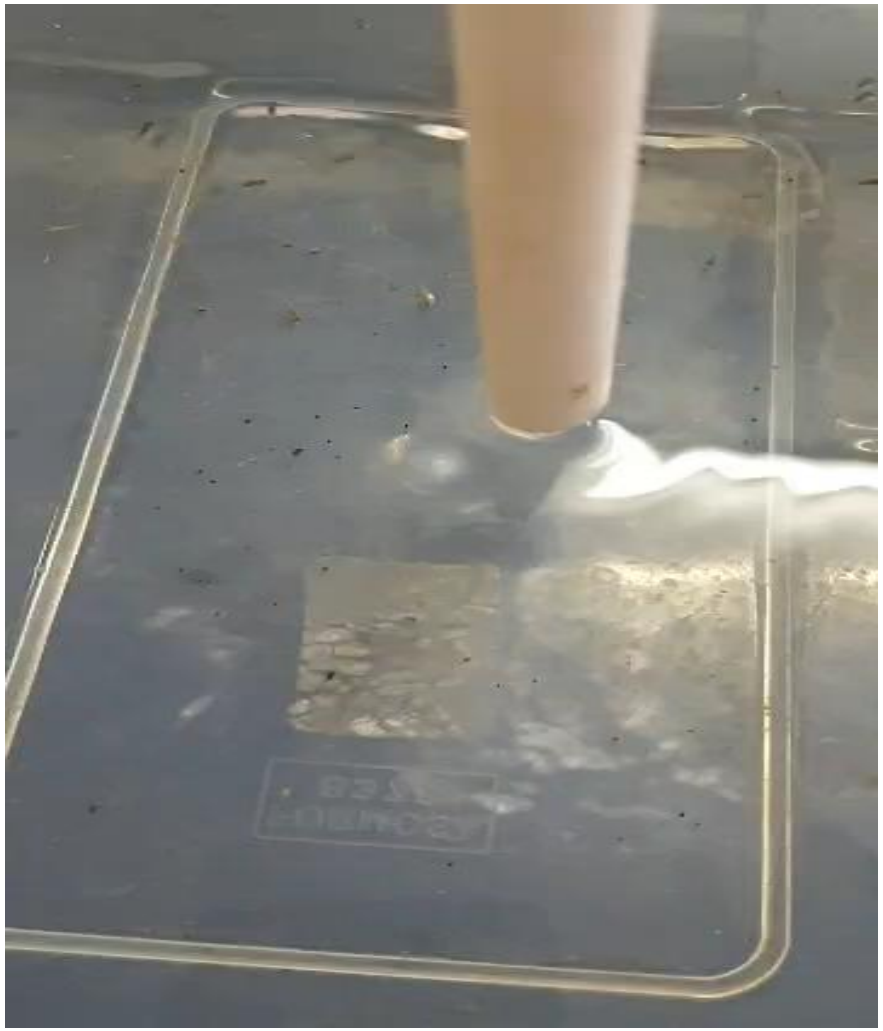


FISH HATCHERIES

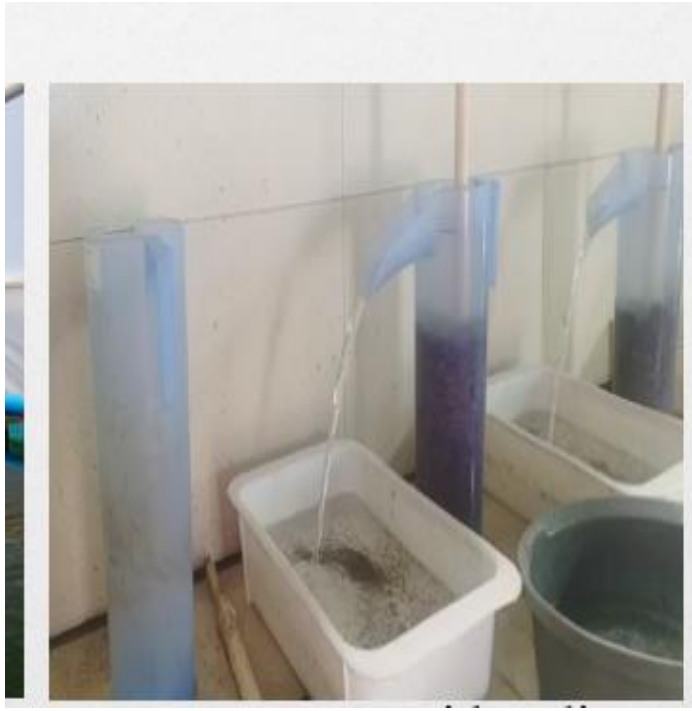
SIMPLE LAYOUT



FISH HATCHERIES







Simple videos

Selected facilities / equipment





Hatching jars

Selected facilities / equipment



Hatching trays

Selected facilities / equipment

- Mostly there are ponds available but even tanks can be used for holding and rearing brood stock
- Breeding or spawning pond, tanks or hapas
- Nursery pond, tanks or hapas
- Water supply system
- Aeration system





Selected facilities / equipment

Pumps (for recirculatory egg incubation system)

Electricity supply and/or generator

Basins, buckets, containers

Seine nets, scoop nets, grading basket

Sensitive scale for weighing fry and fingerlings

Accessories for packing of fry and fingerlings

HATCHERY PONDS

- Hatcheries consist of multiple pond types based on the **size of fish** to be housed and **purpose**.

1. **Nursery pond:** for growing/nursing fry to fingerlings

2. **Brood stock pond:** for rearing/holding the breeders for spawning

3. **Sex-reversal pond**: for sex-reversal of newly hatched fry





HATCHERY PONDS

1. **Conditioning pond:** for holding fingerlings before transport
2. **Grow-out pond:** for growing fingerlings till harvest (to table-size)
3. **Quarantine pond:** for introducing new fish to the farm or for treatment purposes
4. **Reservoir pond:** for storing inlet water before use in the hatchery and nursery

HATCHERY PONDS

HAPAS

You can put broodstock in hapas; collect eggs every week or a fortnight depending on how other parameters.







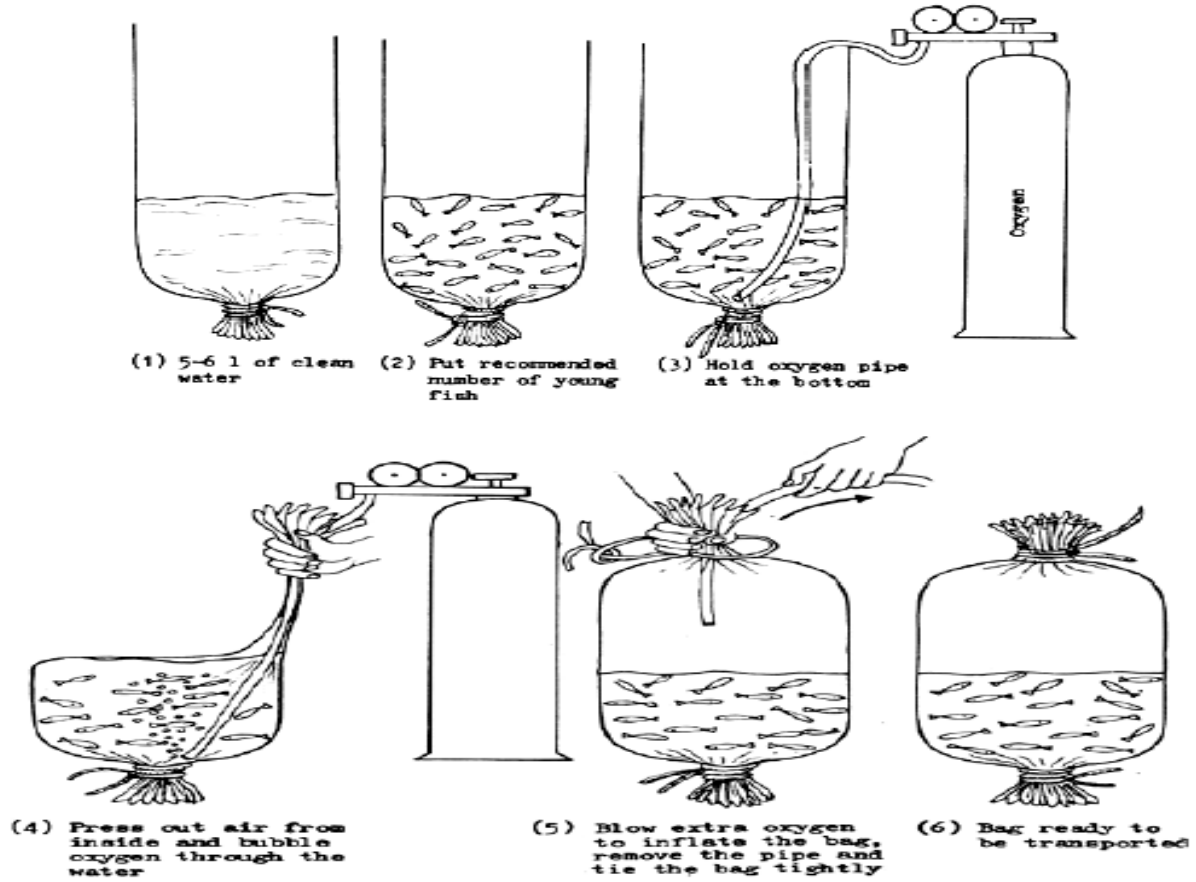
Plate 5. Hapas set up for breeding tilapia at the WorldFish facility in Abbassa, Egypt.

FINGERLING CONDITIONING

- Fingerlings need to be prepared for transport before they are packed;
- They are starved for at least 24hour, placed in clean well oxygenated water;
- The main problems during fish transportation are;
- Shortage of oxygen, high ammonia production from excreta, high temperatures (in hot seasons or areas), and the stress due to handling.

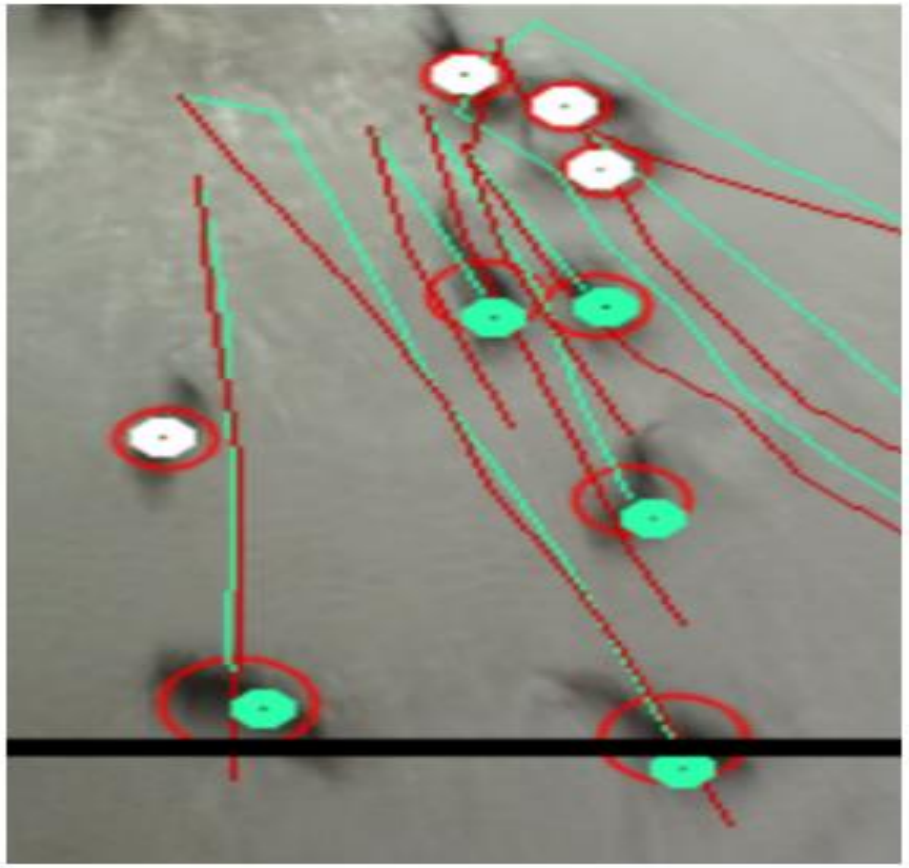
FINGERLING CONDITIONING

- **Condition** the fingerlings before transportation
- Normally 500 to 1,000 fingerling are packed per bag



PACKING

- As counting of all the fry is time consuming and tedious job, three batches of 500 to 1,000 fingerling are counted and weighed;
- The average weight is used to estimate the remaining fingerling and transferred to each bag.



(b) Resulting Image from the Counter

PACKING

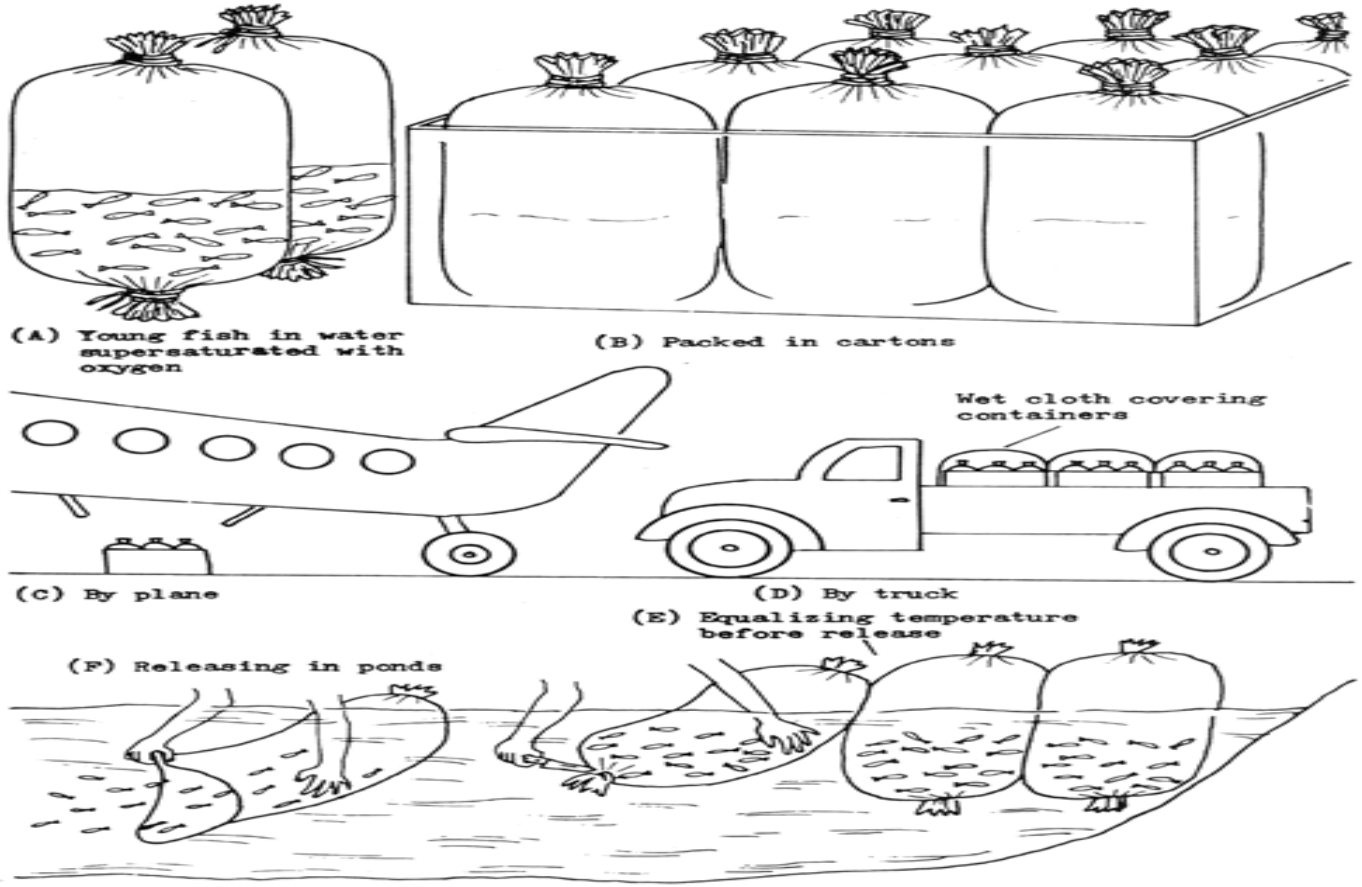




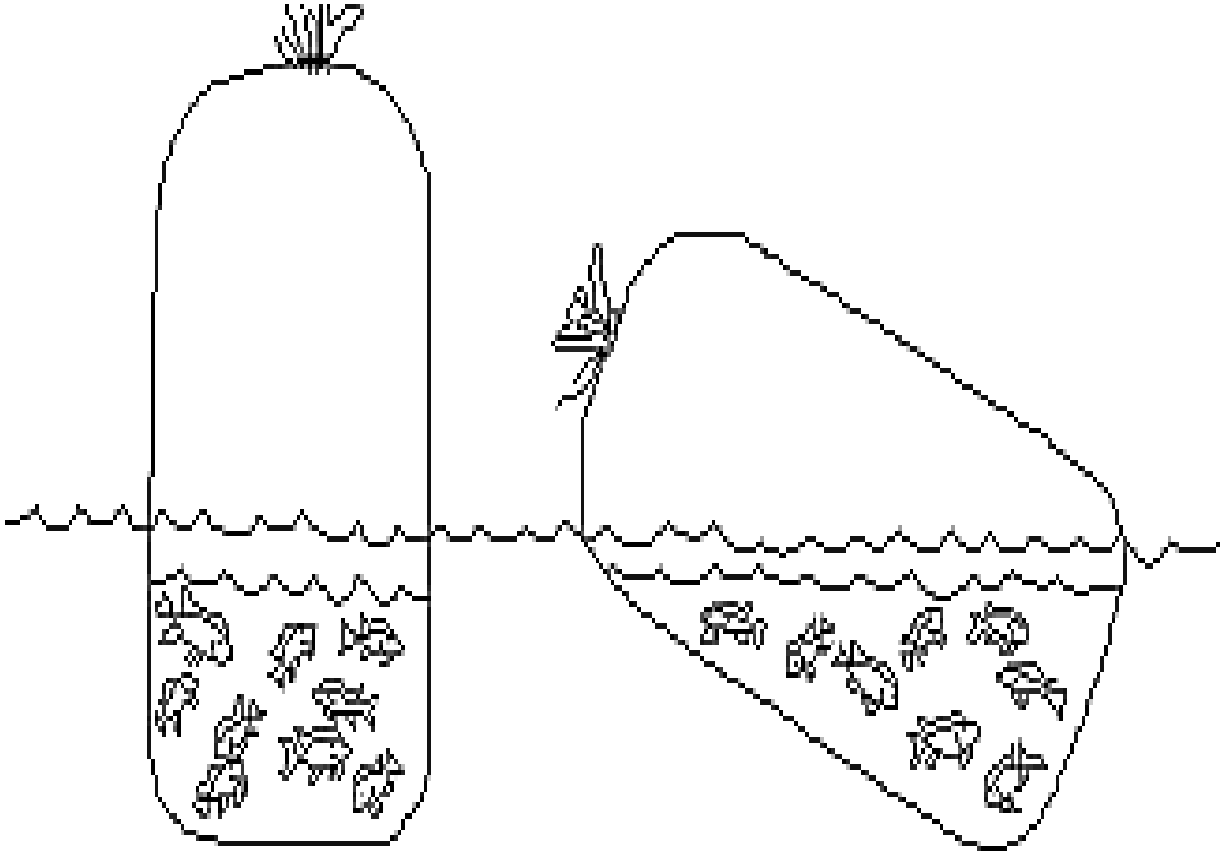


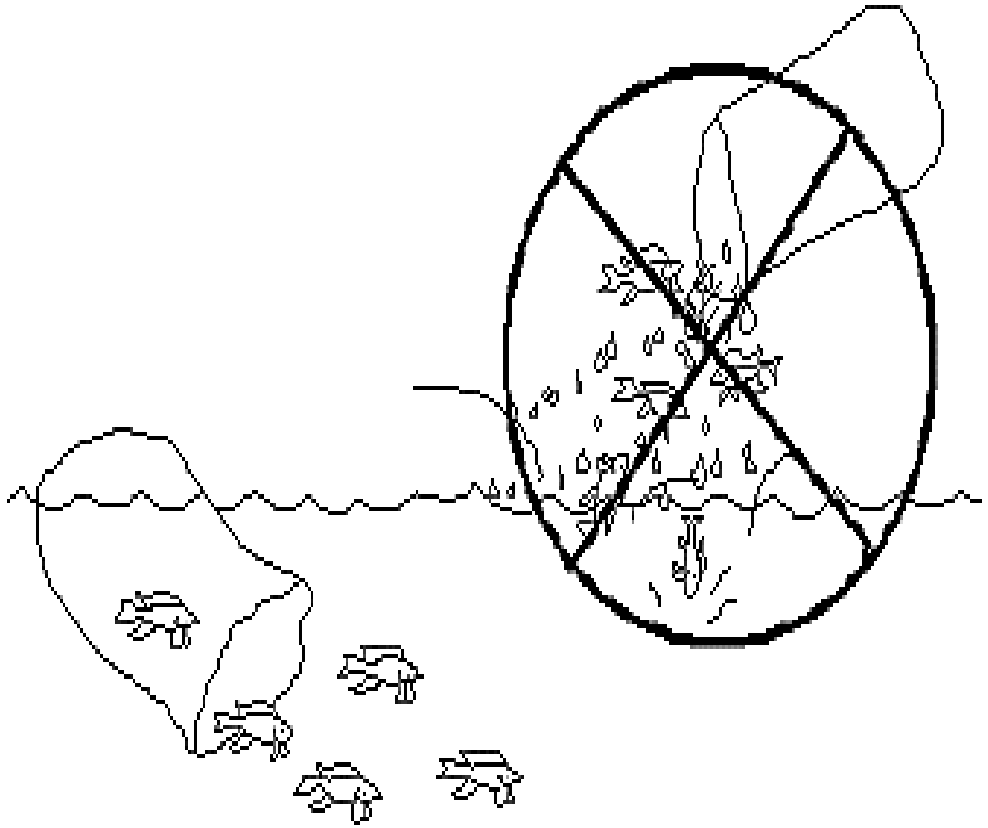
FINGERLING TRANSPORTATION

- Well-conditioned fingerlings packed with clinical class oxygen can be transported to any part of the country up to 12 hours by road or any part of the world by air.



ACCLIMATIZATION





Broodstock management

What makes a good broodstock?

CLASS DISCUSSION

Broodstock management

What makes a good broodstock?

- Normal body shape and colour
- Overall health status
- The best growth and food conversion ratio
- Normal behaviour (quick response to food distribution, fast swimming controlled buoyancy).
- Known source (Credible)
- The age of broodstock should not exceed 5 years

Broodstock management

- ❑ Broodstock selection
  - ✓ Fast growing;
  - ✓ Free from diseases and
  - ✓ No deformities
  - ✓ Free from wounds and other injuries;
  - ✓ Sexually mature and not too old.



Broodstock management

- Stocking density and Sex ratio
  - ✓ 1 to 2 fish per m^2 (1 M: 1F - 6F) in ponds
  - ✓ KMU ponds 1M: 3F
  - ✓ 6 fish per m^2 (3 M: 3F) in hapas;
  - ✓ 10 fish / m^2 (5M: 5F) in tanks.

Broodstock management

- ❑ Broodstock handling;
  - ✓ Use fine seine net, scoops, buckets and basins
- ❑ Broodstock feeding;
  - ✓ 1-3% of body weight.
- ❑ Broodstock conditioning;
  - ✓ Separation of brooders during the cold season.



END

THANK YOU FOR YOUR ATTENTION

